FOR A SMOOTH EQUIDIMENSIONAL SCHEME OVER A BASE OF DIMENSION ONE THE BIVARIANT CHOW RING IS CHOW HOMOLOGY AND COMPOSITION BECOMES INTERSECTION

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FCB (sources/stacks-chow/pinned-original-source.tex line 14979).

Allowed standard prerequisites (editorial).

- The isomorphism between the bivariant ring and the Chow ring via the structure morphism. Same chapter, Proposition labelled proposition-compute-bivariant, cited at proof lines 14992-14996. Supplies the first isomorphism and the description of its inverse.
- Operational Chow cohomology over a base of dimension one. Same chapter, Lemma labelled lemma-chow-cohomology-towards-base-dim-1, cited at proof lines 14994-14996. Supplies the second isomorphism and its inverse.
- The intersection product in the dimension one theory. Same chapter, the definition of the product preceding this lemma, invoked at proof lines 15006-15010. Identifies the composed class with alpha . beta.

Source reference. Original proposition 0FBT: sources/stacks-chow/pinned-original-source.tex, source lines 14003–14003. Same-chapter pointer into the author environment proposition at source line 14003; invoked at proof line 14997 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0FC5: sources/stacks-chow/pinned-original-source.tex, source lines 14734–14734. Same-chapter pointer into the author environment lemma at source line 14734; invoked at proof line 14999 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let (S, δ) be as above. Let X be a smooth scheme over S, equidimensional of dimension d. The map*

$$A^p(X) \longrightarrow \mathrm{CH}_{d-p}(X), \quad c \longmapsto c \cap [X]_d$$

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is an isomorphism. Via this isomorphism composition of bivariant classes turns into the intersection product defined above.

Proof. Denote $g : X \to S$ the structure morphism. The map is the composition of the isomorphisms

$$A^p(X) \to A^{p-d+1}(X \to S) \to \mathrm{CH}_{d-p}(X)$$

The first is the isomorphism $c \mapsto c \circ g^*$ of Proposition 0FBT and the second is the isomorphism $c \mapsto c \cap [S]_1$ of Lemma 0FC5. From the proof of Lemma 0FC5 we see that the inverse to the second arrow sends $\alpha \in \mathrm{CH}_{d-p}(X)$ to the bivariant class c_α which sends $\beta \in \mathrm{CH}_*(Y)$ for Y locally of finite type over k to $\alpha \times \beta$ in $\mathrm{CH}_*(X \times_k Y)$. From the proof of Proposition 0FBT we see the inverse to the first arrow in turn sends c_α to the bivariant class which sends $\beta \in \mathrm{CH}_*(Y)$ for $Y \to X$ locally of finite type to $\Delta^!(\alpha \times \beta) = \alpha \cdot \beta$. From this the final result of the lemma follows. $\square$